

# Changyu Zhao

GRADUATE STUDENT · COMPUTER SCIENCE

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## Education

### Stanford University

Stanford, CA, U.S.A

M.S. COMPUTER SCIENCE (COMPUTER AND NETWORK SECURITY)

Sep. 2025 - June 2027 (expected)

- Selected Courses: Advanced Computer and Network Security, Applied Cryptography, Secure Systems Design, Web Security, Privacy Engineering.

### University of Wisconsin-Madison

Madison, WI, U.S.A

B.S. COMPUTER SCIENCE (HONORS) AND MATHEMATICS

Sep. 2021 - May 2025

- GPA: 3.95/4.00, Dean's List (all semesters).
- Thesis: *Advancing Dynamic Binary Analysis with Hardware Assisted Tracing* (Advisor: Prof. Patrick McDaniel).

## Research Experience

### MadS&P - Security and Privacy Research Group at UW-Madison

Madison, WI, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. PATRICK MCDANIEL

Sep. 2024 - Aug. 2025

- Conducting scientific measurements and evaluations of the performance advantages of **LibIHT** over existing dynamic binary tracing framework, with a focus on the runtime overhead, tracing accuracy, and resilience to anti-tracing techniques.
- Developing a comprehensive evaluation methodology to assess the performance and effectiveness in real-world reverse engineering scenarios, including large-scale softwares and malware samples, to provide insights on practical implications of security research and industry applications.
- Contributing to a research paper to present the findings and insights from the performance evaluation, targeting top-tier security conferences.

### SEFCOM - Laboratory of Security Engineering for Future Computing at ASU

Tempe, AZ, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. YAN SHOSHITAISHVILI

May 2024 - Aug. 2024

- Assisted in the design and development of **Oxidizer**, a Rust-specific decompiler on top of *angr* C decompilation pipeline, aimed at producing Rust pseudo-code with recovered control flow structures, function prototypes, and common-used macros and struct data types from Rust binaries.
- Developed a Rust type scanner in Rust to collect type information from Rust standard libraries and popular crates, and integrated the type information into the decompilation process to recover Rust-specific data types and function signatures with pattern matching and type inference algorithms.
- Collected a comprehensive dataset of Rust binaries with source code to perform evaluations on the decompilation accuracy and effectiveness.

### SEFCOM - Laboratory of Security Engineering for Future Computing at ASU

Tempe, AZ, U.S.A

UNDERGRADUATE RESEARCH ASSISTANT, ADVISOR: PROF. YAN SHOSHITAISHVILI

May 2023 - Jul. 2023

- Engineered **LibHopper**, an automated tool for discovering exploitation primitives of the internal state structures in user libraries.
- Designed analyzation pipeline consisting of debugger scripts to setup context for the internal state structures, and symbolic execution engine to evaluate security implications on memory corruption and control flow hijack, reducing manual analysis time and providing structured assessments.
- Concluded a comprehensive dataset of potential exploitation primitives and possible mitigation strategies in popular user libraries, including *openssl*, *openssh*, *libcurl*, and *libpng*, etc., contributed to the development of automated exploit generation tools and vulnerability scanners.

## Work Experience

### UW-Madison Cybersecurity Operation Center

Madison, WI, U.S.A

CYBERSECURITY STUDENT INTERN ANALYST

Feb. 2023 - May 2025

- Conducted security incident response and analysis for 400+ security incidents, including malware infections, phishing attack, unauthorized access attempts, and network intrusions, utilized Elastic search for log analysis, threat hunting and investigation, and recommending remediation actions.
- Managed monthly identity and access management (IAM) compliance attestation audits for 500+ employee accounts.

### Tencent Security Xuanwu Lab

Beijing, China

SECURITY RESEARCH INTERN

Jul. 2023 - Aug. 2023

- Engineered **LibIHT**, a cross platform dynamic binary tracing framework in C, C++, and Python to analyze control flow behavior of large programs with low overhead, leveraging Intel hardware tracing features such as Last Branch Record (LBR) and Branch Trace Store (BTS).
- Architected cross platform capabilities for Linux and Windows kernel driver to interface with Intel hardware features and provide full control on user-space tracing, implemented a unified set of API abstraction on kernel APIs for cross platform compatibility and avoiding duplicated code logic.
- Developed a user-space library to provide user-friendly high-level API access for customized user applications to interact with the tracing framework, implemented a plugin system for easy integration with debuggers like GDB and WinDbg to visualize and analyze the tracing data.

## Honors & Awards

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|---------|------------------------------------------------------------------------------|-----------------------------|
| 2022-25 | <b>Finalist</b> , DEFCON CTF Hacking Competition World Final                 | <i>Las Vegas, NV, U.S.A</i> |
| 2021-25 | <b>Dean's List</b> , UW-Madison College of Letters & Science                 | <i>Madison, WI, U.S.A</i>   |
| 2024    | <b>David Dewitt Undergraduate Scholarship</b> , UW-Madison Computer Sciences | <i>Madison, WI, U.S.A</i>   |
| 2023    | <b>Finalist</b> , CSAW'23 CTF Final (US-Canada region)                       | <i>New York, NY, U.S.A</i>  |
| 2022    | <b>49/3658, Top 2%</b> , Cyber Skyline National Cyber League Team Game       | <i>Madison, WI, U.S.A</i>   |

## Publication

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- Yibo Liu, Zion Basque, Arvind Raj, Chavin Udomwongsa, Chang Zhu, Jie Hu, **Changyu Zhao**, Fangzhou Dong, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, and Ruoyu Wang. "Toward Concise and High-fidelity Rust Decompilation." In *Proceedings of the Network and Distributed System Security Symposium (NDSS)*, February 2026. (In submission)
- **Changyu Zhao**, Yohan Beugin, Jean-Charles Noirot Ferrand, Quinn Burke, Guancheng Li, and Patrick McDaniel. "LibIHT: A Hardware-Based Approach to Efficient and Evasion-Resilient Dynamic Binary Analysis". In *Proceedings of the 1st Workshop on Software Understanding and Reverse Engineering (SURE 2025)*. ACM, October 2025.

## Presentation

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### BlackHat USA

*Las Vegas, NV, U.S.A*

ARSENAL PRESENTER FOR "LIBIHT: A CROSS-PLATFORM LIBRARY FOR ACCESSING INTEL HARDWARE TRACE FEATURES"

*Aug. 2024*

- Presented **LibIHT** to industry professionals, highlighting its advanced tracing capabilities in dynamic control flow analysis on large programs.
- Discussed use cases for LibIHT in vulnerability research, showcasing its potential to improve analysis capabilities for reverse engineering tasks.

## Extracurricular Activity

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### r3kapig CTF Team

*International*

TEAM MEMBER

*Sep. 2021 - Present*

- Competed in over 40 high-quality 48-hour Capture The Flag (CTF) competitions. Expert in **binary exploitation** and **reverse engineering**.
- Entered DEFCON CTF world final competitions from 2022 to 2025, ranked 12th, 7th, 4th, and 5th respectively.

### Cybersecurity UW Club

*Madison, WI, U.S.A*

CORE CTF TEAM MEMBER, LECTURER

*Sep. 2021 - May 2025*

- Led the Cybersecurity UW CTF team to **top 2%** in a national cybersecurity competition, providing strategic guidance and technical expertise.
- Delivered over 10 workshop lectures on **binary exploitation** and **reverse engineering**, contributing to the skill development of 30+ club members.
- Deepened technical skills while contributing to the broader cybersecurity community through leadership, mentoring, and knowledge sharing.

### pwn.college

*Online*

INDEPENDENT LEARNER, VOLUNTEER PEER MENTOR

*Dec. 2021 - Present*

- Attained **Yellow Belt** and **Blue Belt** certifications by completing all challenges in the pwn.college, currently ranked **43 out of 2500+** participants.
- Developed comprehensive skills in software and system security, including reverse engineering, software exploitation, and kernel security.
- Mentored and guided students, particularly from underrepresented groups, by helping them build foundational skills in cybersecurity.
- Expanded access to cybersecurity education by actively participating in the learning community, answering questions, and offering advice.

## Skills

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**Programming** Python, C, C++, Assembly (x86, x64, ARM), Rust, Java, JavaScript, Go, Bash, SQL

**Security Tools** IDA Pro, Ghidra, GDB, Windbg, OllyDbg, QEMU, AFL, LLVM, Unicorn, Frida, Burp Suite, Wireshark, Metasploit, Kali Linux

**Languages** Chinese (Native), English (Fluent)